



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA1017 – SOFTWARE ENGINEERING

INDIVIDUAL REFLECTION REPORT

INTELLIGENT DISASTER EARLY WARNING AND EVACUATION MANAGEMENT SYSTEM

NAME: DAVIS DAMIEN M

REGISTER NO.: 192521124

ROLE: Risk Assessment and Algorithm Design

DESIGN / DEVELOPMENT DECISION: I contributed to the risk-assessment logic, alert-priority rules, algorithm documentation, and pseudocode. I selected a transparent rule-based risk method for the MVP because the project uses simulated data and does not contain a sufficiently large historical dataset for a validated machine-learning model.

CHALLENGE FACED AND SOLUTION: The major challenge was representing AI and ML concepts without exaggerating the capability of the prototype. I addressed this by showing contributing factors, using clear risk categories such as Normal, Watch, Warning, and Critical, and requiring human verification before an alert is treated as official.

LEARNING GAINED: This assignment strengthened my understanding of algorithm design, data quality, explainable decision support, and responsible AI. I learned the difference between a demonstration score and a scientifically validated prediction model. I also understood why confidence, uncertainty, data provenance, and human accountability are important in emergency systems.

SDG RELEVANCE: The system supports SDG 13 by demonstrating how risk information can contribute to preparedness and climate-related hazard awareness, while avoiding unsupported claims of measured real-world impact.

COURSE OUTCOME CONTRIBUTION: My work supports the course outcome related to algorithms and AI/ML application by connecting hazard indicators and operational conditions to explainable risk-assessment logic.

POSSIBLE IMPROVEMENT: A future version should use authoritative historical datasets, model calibration, accuracy evaluation, bias analysis, and independent validation.